Optimal Portfolios

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Review of HF return modeling

- LOB data structure
  - Bid/Ask Price/Volume, deeper levels
- Sampling frequency
  - Trading is not continuous.
- Different clocks
  - calendar, transaction, volume
- Universality
  - OOS, unseen stocks, testing periods, path dependency

Long-Short Spread Portfolio

Definition: A strategy that takes long positions in assets expected to outperform and short positions in assets expected to underperform.

- Constructed by ranking stocks based on a characteristic (e.g., momentum, value).
- Returns are computed as the spread between long and short positions.
- to reduce broad market exposure.

Example:

- Momentum: Long high past return stocks, short low past return stocks.
- ML: Long stocks with highest forecasts, short those with lowest forecasts

Mean Variance Portfolio

Mean Variance Portfolio Implementation

Integrated Estimation and Optimization

Maximum Sharpe Ratio Regression

Considering trading costs

Basic properties of portfolio returns

- Portfolio: n-vector of weights indicating the fraction of portfolio wealth held in each asset:

$$\boldsymbol{w} = (w_1, \cdots, w_n)^\top : \sum_{i=1}^{n} w_i = 1$$

- Portfolio return: $R_{\boldsymbol{w}} = \boldsymbol{w}^\top \boldsymbol{R} = \sum_{i=1}^{n} w_i R_i$
- Expected portfolio return $\mu_{\boldsymbol{w}} := \mathbb{E}[R_{\boldsymbol{w}}] = \boldsymbol{w}^\top \boldsymbol{\mu} = \sum_{i=1}^{n} w_i \mu_i$
- Portfolio variance: $\sigma_{\boldsymbol{w}}^2 = \mathrm{Var}[R_{\boldsymbol{w}}] = \mathbb{E}[(R_{\boldsymbol{w}} - \mu_{\boldsymbol{w}})^2] = \boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w} = \sum_{i=1}^{n} \sum_{j=1}^{n} w_i w_j \sigma_i \sigma_j \rho_{ij}$ where

$$\boldsymbol{\Sigma} = \begin{pmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n1} & \sigma_{n2} & \cdots & \sigma_{nn} \end{pmatrix}$$

is the covariance matrix of returns with $\sigma_{ii} = \sigma_i^2$

Warm-up Example

Assume we have two risky assets. Asset 1 has an expected return of 0.1 with a variance of 0.04, and asset 2 has an expected return 0.15 with a variance of 0.09. The covariance between the returns is 0.05.

Question: Calculate the portfolio expected return and variance for the weights $(0.3, 0.7)$.

Warm-up Example

Assume we have two risky assets. Asset 1 has an expected return of 0.1 with a variance of 0.04, and asset 2 has an expected return 0.15 with a variance of 0.09. The covariance between the returns is 0.05.

Question: Calculate the portfolio expected return and variance for the weights $(0.3, 0.7)$.

Answer:

$$\boldsymbol{\mu} = \begin{pmatrix} 0.1 \\ 0.15 \end{pmatrix}, \boldsymbol{\Sigma} = \begin{pmatrix} 0.04 & 0.05 \\ 0.05 & 0.09 \end{pmatrix} \quad \boldsymbol{w} = (0.3, 0.7)^\top$$

$$E[R_{\boldsymbol{w}}] = \boldsymbol{w}^\top \boldsymbol{\mu} = 0.1 \times 0.3 + 0.15 \times 0.7 = 0.135$$

$$\mathrm{Var}[R_{\boldsymbol{w}}] = \boldsymbol{w}^\top \Sigma \boldsymbol{w} = \begin{pmatrix} 0.3 & 0.7 \end{pmatrix} \begin{pmatrix} 0.04 & 0.05 \\ 0.05 & 0.09 \end{pmatrix} \begin{pmatrix} 0.3 \\ 0.7 \end{pmatrix}$$

$$= \begin{pmatrix} 0.047 & 0.078 \end{pmatrix} \begin{pmatrix} 0.3 \\ 0.7 \end{pmatrix} = 0.0687$$

Markowitz Mean-variance Optimization with Risky Assets

Goal: Given n risk assets with $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ such that $\sigma_{ii} > 0$ $(i = 1, 2, \cdots, n)$, evaluate different portfolio $\boldsymbol{w} = (w_1, \cdots, w_n)^\top$ using the mean-variance pair of the portfolio $(\mu_{\boldsymbol{w}}, \sigma^2_{\boldsymbol{w}})$ with preferences for

- higher expected returns $\mu_{\boldsymbol{w}}$
- lower variance $\sigma^2_{\boldsymbol{w}}$

Formulation I: Risk minimization: For a given choice of target mean return $\overline{\mu}$, choose the portfolio $\boldsymbol{w}$ to

$$\begin{aligned} \text{Minimize}: \quad & \tfrac{1}{2}\boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w} \\ \text{Subject to}: \quad & \boldsymbol{w}^\top \boldsymbol{\mu} = \overline{\mu} \\ & \boldsymbol{w}^\top \mathbf{1}_n = 1 \end{aligned}$$

where $\mathbf{1}_n = (1, 1, \cdots, 1)^\top \in \mathbb{R}^n$

Solution: Applying the method of Lagrange multipliers to the convex optimization (minimization) problem subject to linear constraints.

Lagrange Multipliers

The Lagrange multipliers is a strategy for finding the local (or global) maxima and minima of a function subject to equality constraints

- The basic idea is to convert a constrained problem into an unconstrained problem so that the first-order condition (derivative test) can still be applied

Original (constrained) problem:

$$\begin{aligned} \text{Minimize}: &\quad f(\boldsymbol{x}) \text{ with } \boldsymbol{x} \in \mathbb{R}^n \\ \text{Subject to}: &\quad g_j(\boldsymbol{x}) = c_j \ (j = 1, 2, \cdots, m) \end{aligned}$$

Lagrangian function:

$$\mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda}) = f(\boldsymbol{x}) + \sum_{j=1}^{n} \lambda_j (c_j - g_j(\boldsymbol{x}))$$

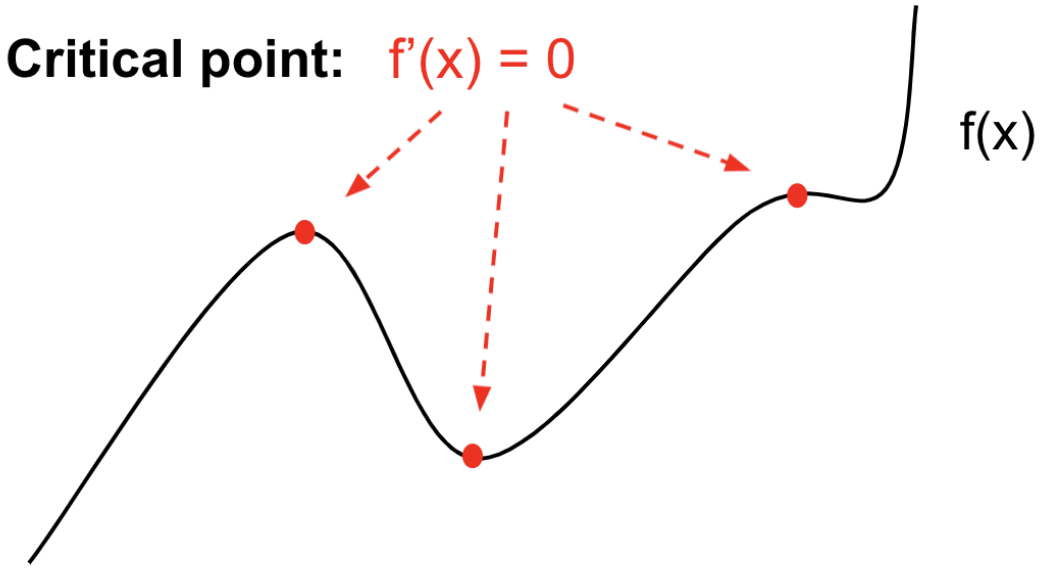
- $\mathcal{L} : \mathbb{R}^{n+m} \to \mathbb{R}$

Lagrange Multipliers

Translated (unconstrained) optimization problem:

$$\begin{cases} \frac{\partial}{\partial x_i}\mathcal{L}(\boldsymbol{x},\boldsymbol{\lambda}) = \frac{\partial}{\partial x_i}f(\boldsymbol{x}) - \sum_{i=1}^{m}\lambda_i\frac{\partial}{\partial x_i}g_i(\boldsymbol{x}) = \mathbf{0},\ (i=1,2,\cdots,n) \\ \frac{\partial}{\partial \lambda_j}\mathcal{L}(\boldsymbol{x},\boldsymbol{\lambda}) = c_j - g_j(\boldsymbol{x}) = 0,\ (j=1,2,\cdots,m) \end{cases}$$

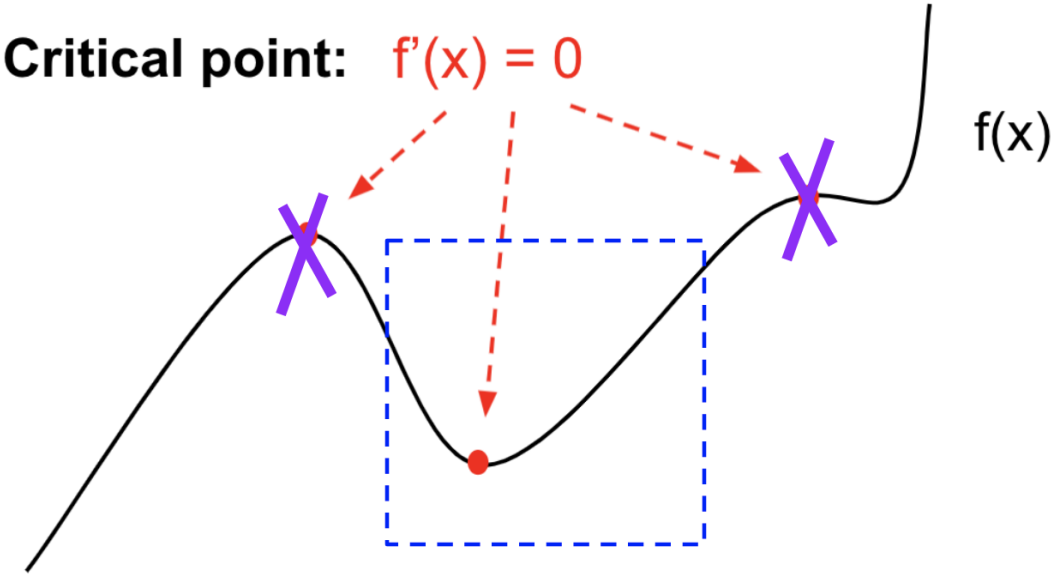
- (m+n) unknowns and (m+n) equation*s
- Any solution of this system $(\boldsymbol{x}_*;\boldsymbol{\lambda}_*)$ is called a critical point of the Lagrange function $\mathcal{L}$



Method Of Lagrange Multipliers

Sufficient condition for global extrema: Let $(\boldsymbol{x}_*, \boldsymbol{\lambda}_*)$ be a critical point of $\mathcal{L}$ of optimization problem, If $\mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda}_*)$ is convex in $\boldsymbol{x}$, then $\boldsymbol{x}_*$ is a global minimum of $f(\boldsymbol{x})$ subject to $g_j(\boldsymbol{x}) = c_j$ $(j = 1, 2, \cdots, m)$.

- ⋆ This sufficient condition is enough for the mean-variance optimization problem!



Risk Minimization Without Budget Constraint

$$\begin{aligned} \text{Minimize}: \quad & \tfrac{1}{2}\boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w} \\ \text{Subject to}: \quad & \boldsymbol{w}^\top \boldsymbol{\mu} = \overline{\mu} \end{aligned}$$

Risk Minimization Without Budget Constraint

$$\begin{aligned} \text{Minimize}: \quad & \tfrac{1}{2}\boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w} \\ \text{Subject to}: \quad & \boldsymbol{w}^\top \boldsymbol{\mu} = \overline{\mu} \end{aligned}$$

- Define the Lagrangian $\mathcal{L}(\boldsymbol{w}, \lambda_1) = \frac{1}{2}\boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w} + \lambda_1(\overline{\mu} - \boldsymbol{w}^\top \boldsymbol{\mu})$
- Derive the first-order conditions

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \boldsymbol{w}} &= \boldsymbol{0}_n = \boldsymbol{\Sigma} \boldsymbol{w} - \lambda_1 \boldsymbol{\mu} \\ \frac{\partial \mathcal{L}}{\partial \lambda_1} &= 0 = \overline{\mu} - \boldsymbol{w}^\top \boldsymbol{\mu} \end{aligned}$$

- Solve for $\boldsymbol{w}$ in terms of λ_1:

$$\boldsymbol{w}_* = \lambda_1 \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}$$

Final Solution for Portfolio Weights

Using the return constraint $\boldsymbol{w}^\top \boldsymbol{\mu} = \overline{\mu}$:

$$(\lambda_1 \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu})^\top \boldsymbol{\mu} = \overline{\mu}$$

This simplifies to:

$$\lambda_1 \boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu} = \overline{\mu}$$

Thus, the Lagrange multiplier λ_1 is:

$$\lambda_1 = \frac{\overline{\mu}}{\boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}}$$

Substituting λ_1 back into the equation* for $\boldsymbol{w}$,

$$\boldsymbol{w}_* = \frac{\overline{\mu}}{\boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}} \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}$$

- The portfolio weights are proportional to $\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$, which reflects the trade-off between risk (via $\boldsymbol{\Sigma}^{-1}$) and expected return (via $\boldsymbol{\mu}$).
- The proportionality factor $\frac{\overline{\mu}}{\boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}}$ ensures that the portfolio achieves the target return $\overline{\mu}$.

Risk Minimization Problem

$$\begin{aligned} \text{Minimize}: &\quad \tfrac{1}{2}\boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w} \\ \text{Subject to}: &\quad \boldsymbol{w}^\top \boldsymbol{\mu} = \overline{\mu}, \boldsymbol{w}^\top \mathbf{1}_n = 1 \end{aligned}$$

- Define the Lagrangian
  $\mathcal{L}(\boldsymbol{w}, \lambda_1, \lambda_2) = \frac{1}{2}\boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w} + \lambda_1(\overline{\mu} - \boldsymbol{w}^\top \boldsymbol{\mu}) + \lambda_2(1 - \boldsymbol{w}^\top \mathbf{1})$
- Derive the first-order conditions

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \boldsymbol{w}} &= \mathbf{0}_n = \boldsymbol{\Sigma} \boldsymbol{w} - \lambda_1 \boldsymbol{\mu} - \lambda_2 \mathbf{1}_n \\ \frac{\partial \mathcal{L}}{\partial \lambda_1} &= 0 = \overline{\mu} - \boldsymbol{w}^\top \boldsymbol{\mu} \\ \frac{\partial \mathcal{L}}{\partial \lambda_2} &= 0 = 1 - \boldsymbol{w}^\top \mathbf{1}_n \end{aligned}$$

- Solve for $\boldsymbol{w}$ in terms of λ_1, λ_2:

$$\boldsymbol{w}_* = \lambda_1 \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu} + \lambda_2 \boldsymbol{\Sigma}^{-1} \mathbf{1}_n$$

Risk Minimization Problem

- Solve for λ_1 and λ_2 by substituting for $\boldsymbol{w}^*$:

$$\begin{aligned}
\overline{\mu} &= \boldsymbol{w}_*^\top \boldsymbol{\mu} = \lambda_1(\boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}) + \lambda_2(\boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1}\mathbf{1}_n) \\
1 &= \boldsymbol{w}_*^\top \mathbf{1}_n = \lambda_1(\boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1}\mathbf{1}_n) + \lambda_2(\mathbf{1}_n^\top \boldsymbol{\Sigma}^{-1}\mathbf{1}_n) \\
\Rightarrow \quad & \begin{pmatrix} \overline{\mu} \\ 1 \end{pmatrix} = \begin{pmatrix} a & b \\ b & c \end{pmatrix} \begin{pmatrix} \lambda_1 \\ \lambda_2 \end{pmatrix} \text{ with}
\end{aligned}$$

$a = \boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$, $b = \boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1}\mathbf{1}_n$, and $c = \mathbf{1}_n^\top \boldsymbol{\Sigma}^{-1}\mathbf{1}_n$.
Hence

$$\begin{pmatrix} \lambda_1 \\ \lambda_2 \end{pmatrix} = \begin{pmatrix} a & b \\ b & c \end{pmatrix}^{-1} \begin{pmatrix} \overline{\mu} \\ 1 \end{pmatrix}.$$

Risk Minimization Problem

Variance of optimal portfolio with return $\overline{\mu}$: With the given values of λ_1 and λ_2, the solution portfolio $\boldsymbol{w}_* = \lambda_1 \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} + \lambda_2 \boldsymbol{\Sigma}^{-1}\mathbf{1}_n$ has minimum variance equal to

$$\begin{aligned}
\sigma_*^2 &= \boldsymbol{w}_*^\top \boldsymbol{\Sigma} \boldsymbol{w}_* \\
&= \lambda_1^2(\boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}) + 2\lambda_1\lambda_2(\boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1}\mathbf{1}_n) + \lambda_1^2(\mathbf{1}_n^\top \boldsymbol{\Sigma}^{-1}\mathbf{1}_n) \\
&= (\lambda_1, \lambda_2) \begin{pmatrix} a & b \\ b & c \end{pmatrix} \begin{pmatrix} \lambda_1 \\ \lambda_2 \end{pmatrix}
\end{aligned}$$

Substituting $\begin{pmatrix} \lambda_1 \\ \lambda_2 \end{pmatrix} = \begin{pmatrix} a & b \\ b & c \end{pmatrix}^{-1} \begin{pmatrix} \overline{\mu} \\ 1 \end{pmatrix}$ gives

$$\boxed{\sigma_*^2 = (\overline{\mu}, 1) \begin{pmatrix} a & b \\ b & c \end{pmatrix}^{-1} \begin{pmatrix} \overline{\mu} \\ 1 \end{pmatrix} = \frac{1}{ac - b^2}(c\overline{\mu}^2 - 2b\overline{\mu} + a)}$$

- Optimal portfolio has variance σ_*^2: parabolic in the mean

Other Formulations

Formulation II: Expected Return Maximization. For a given choice of target return variance $\overline{\sigma}^2$, choose the portfolio $\boldsymbol{w}$ to

$$\begin{aligned} \text{Maximize}: \quad & \boldsymbol{w}^\top \boldsymbol{\mu} \\ \text{Subject to}: \quad & \tfrac{1}{2}\boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w} = \overline{\sigma}^2 \\ & \boldsymbol{w}^\top \mathbf{1}_n = 1 \end{aligned}$$

Formulation III: Risk Aversion Optimization. Let $\lambda \geq 0$ denote the risk aversion preference gauging the trade-off between risk and return. Choose the portfolio $\boldsymbol{w}$ to

$$\begin{aligned} \text{Maximize}: \quad & \mathbb{E}[R_{\boldsymbol{w}}] - \tfrac{\lambda}{2}\mathrm{Var}[R_{\boldsymbol{w}}] = \boldsymbol{w}^\top \boldsymbol{\mu} - \tfrac{\lambda}{2}\boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w} \\ \text{Subject to}: \quad & \boldsymbol{w}^\top \mathbf{1}_n = 1 \end{aligned}$$

Mean-variance with (Additional) Risk-free Asset

Risk-free asset: In addition to the risky asset ($i = 1, 2, \cdots, n$), we assume there is a risk-free asset ($i = 0$) for which

$$R_0 = \mu_0, \text{ i.e., } \mathbb{E}[R_0] = \mu_0, \ \mathrm{Var}[R_0] = 0$$

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Portfolio with investment in risk-free asset

- Suppose the investor can invest in the n risky investment as well as in the risk-free asset: $\boldsymbol{w}^\top \mathbf{1}_n = \sum_{i=1}^n w_i$ is invested in risky assets and $1 - \boldsymbol{w}^\top \mathbf{1}_n$ is invested in the risk-free asset.
- If borrowing is allowed, $(1 - \boldsymbol{w}^\top \mathbf{1}_n)$ can be negative
- Portfolio: $R_{\boldsymbol{w}} = \boldsymbol{w}^\top \boldsymbol{R} + (1 - \boldsymbol{w}^\top \mathbf{1}_n) R_0$, where $\boldsymbol{R} = (R_1, \cdots, R_n)$, has expected return and variance

$$\begin{aligned} \mu_{\boldsymbol{w}} &= \boldsymbol{w}^\top \boldsymbol{\mu} + (1 - \boldsymbol{w}^\top \mathbf{1}_n)\mu_0 \\ \sigma_{\boldsymbol{w}}^2 &= \boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w} \end{aligned}$$

Note: R_0 has zero variance and is uncorrelated with $\boldsymbol{R}$.

Mean-variance Optimization with Risk-free Asset

Formulation I′: Risk optimization with risk-free asset For a given choice of target mean return $\overline{\mu}$, choose the portfolio $\boldsymbol{w}$ to

$$\begin{aligned} \text{Minimize}: & \quad \tfrac{1}{2}\boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w} \\ \text{Subject to}: & \quad \boldsymbol{w}^\top \boldsymbol{\mu} + (1 - \boldsymbol{w}^\top \mathbf{1}_n)\mu_0 = \overline{\mu} \end{aligned}$$

Solution: Apply the method of Lagrange multipliers to the convex optimization (minimization):

- Define the Lagrangian

$$\mathcal{L}(\boldsymbol{w}, \lambda_1) = \frac{1}{2}\boldsymbol{w}\boldsymbol{\Sigma}\boldsymbol{w} + \lambda_1[(\overline{\mu} - \mu_0) - \boldsymbol{w}^\top(\boldsymbol{\mu} - \mu_0 \mathbf{1}_n)]$$

- Derive the first-order conditions

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \boldsymbol{w}} &= \mathbf{0}_n = \boldsymbol{\Sigma}\boldsymbol{w} - \lambda_1[\boldsymbol{\mu} - \mathbf{1}_n \mu_0] \\ \frac{\partial \mathcal{L}}{\partial \lambda_1} &= 0 = (\overline{\mu} - \mu_0) - \boldsymbol{w}^\top(\boldsymbol{\mu} - \mathbf{1}_n \mu_0) \end{aligned}$$

Mean-variance Optimization with Risk-free Asset

Solution (cont):

- Solve for $\boldsymbol{w}$ in terms of λ_1:

$$\boldsymbol{w}_* = \lambda_1 \boldsymbol{\Sigma}^{-1}[\boldsymbol{\mu} - \mathbf{1}_n \mu_0]$$

and

$$\lambda_1 = \frac{\overline{\mu} - \mu_0}{(\boldsymbol{\mu} - \mathbf{1}_n \mu_0)^\top \boldsymbol{\Sigma}^{-1} (\boldsymbol{\mu} - \mathbf{1}_n \mu_0)}$$

Mean-variance Optimization with Risk-free Asset

Exercise. Asset 1 has an expected return of 10% with a standard deviation of 25%, and asset 2 has an expected return of 15% and a standard deviation of 35%. The covariance between the returns is 0.0175 and the risk-free rate is 8%.

(a) What is the optimal portfolio consisting of risky assets and risk-free asset if you want an average return of 0.10?

Mean-variance Optimization with Risk-free Asset

Exercise. Asset 1 has an expected return of 10% with a standard deviation of 25%, and asset 2 has an expected return of 15% and a standard deviation of 35%. The covariance between the returns is 0.0175 and the risk-free rate is 8%.

(a) What is the optimal portfolio consisting of risky assets and risk-free asset if you want an average return of 0.10?

Solution: Note that

$$\boldsymbol{\Sigma} = \begin{pmatrix} 0.0625 & 0.0175 \\ 0.0175 & 0.1225 \end{pmatrix}, \boldsymbol{\mu} = \begin{pmatrix} 0.1 \\ 0.15 \end{pmatrix}, \overline{\mu} = 0.1, \text{ and } \mu_0 = 0.08,$$

and we have the formula

$$\boldsymbol{w}_* = \frac{(\overline{\mu} - \mu_0)\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - \mathbf{1}_n\mu_0)}{(\boldsymbol{\mu} - \mathbf{1}_n\mu_0)^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - \mathbf{1}_n\mu_0)}. \tag{1}$$

First, we have

$$\boldsymbol{\Sigma}^{-1} = \frac{1}{0.0625 \times 0.1225 - 0.0175 \times 0.0175} \begin{pmatrix} 0.1225 & -0.0175 \\ -0.0175 & 0.0625 \end{pmatrix}$$

Mean-variance Optimization with Risk-free Asset

Solution (cont I): and

$$\boldsymbol{\mu} - \mathbf{1}_n \mu_0 = \begin{pmatrix} 0.1 \\ 0.15 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \end{pmatrix} \times 0.08 = \begin{pmatrix} 0.02 \\ 0.07 \end{pmatrix}. \text{ Then}$$

$$\begin{aligned} & (\overline{\mu} - \mu_0)\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - \mathbf{1}_n \mu_0) = \\ & \frac{(0.1 - 0.08)}{0.0625 \times 0.1225 - 0.0175 \times 0.0175} \begin{pmatrix} 0.1225 & -0.0175 \\ -0.0175 & 0.0625 \end{pmatrix} \begin{pmatrix} 0.02 \\ 0.07 \end{pmatrix} \\ = \quad & 2.7210884 \times \begin{pmatrix} 0.001225 \\ 0.004025 \end{pmatrix} \end{aligned}$$

and

$$\begin{aligned} & (\boldsymbol{\mu} - \mathbf{1}_n \mu_0)^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - \mathbf{1}_n \mu_0) = \frac{1}{0.0625 \times 0.1225 - 0.0175 \times 0.0175} \times \\ & (0.02, 0.07) \begin{pmatrix} 0.1225 & -0.0175 \\ -0.0175 & 0.0625 \end{pmatrix} \begin{pmatrix} 0.02 \\ 0.07 \end{pmatrix} = 0.04166 \end{aligned}$$

Mean-variance Optimization with Risk-free Asset

Solution (cont II):

Hence we have

$$\boldsymbol{w}_* = \frac{(\overline{\mu} - \mu_0)\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - \mathbf{1}_n\mu_0)}{(\boldsymbol{\mu} - \mathbf{1}_n\mu_0)^\top\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - \mathbf{1}_n\mu_0)} = \frac{2.7210884}{0.04166}\begin{pmatrix}0.001225\\0.004025\end{pmatrix} = \begin{pmatrix}0.08\\0.26\end{pmatrix},$$

and $w_0 = 1 - 0.08 - 0.26 = 0.66$ is the proportion invested in the risk-free asset.

Global Minimum Variance Portfolio (GMVP)

The GMVP focuses solely on minimizing portfolio risk, completely ignoring expected returns. It is formulated as the following convex quadratic program:

$$\begin{array}{ll} \underset{\mathbf{w}}{\text{minimize}} & \mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w} \\ \text{subject to} & \mathbf{1}^T \mathbf{w} = 1 \end{array}$$

Global Minimum Variance Portfolio (GMVP)

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The optimal solution to this problem is given by:

$$\mathbf{w}_{\text{GMVP}} = \frac{1}{\mathbf{1}^T \boldsymbol{\Sigma}^{-1} \mathbf{1}} \boldsymbol{\Sigma}^{-1} \mathbf{1}$$

Due to its simplicity, the GMVP is widely used in academic research to evaluate and compare different covariance matrix estimators, while *avoiding the complexities of estimating expected returns*.

Practical constraints 1

Capital Budget Constraint: Ensures full capital allocation.

$$\mathbf{1}^T \mathbf{w} = 1$$

Long-Only Constraint: No short selling allowed.

$$\mathbf{w} \geq 0$$

Dollar-Neutral (Self-Financing) Constraint: No additional capital investment.

$$\mathbf{1}^T \mathbf{w} = 0$$

Holding Constraint: Restricts asset positions within limits.

$$\mathbf{l} \leq \mathbf{w} \leq \mathbf{u}$$

Practical constraints 2

Leverage Constraint: Limits total exposure to leverage limit L.

$$\|\mathbf{w}\|_1 \leq L$$

Cardinality Constraint: Restricts the number of selected assets.

$$\|\mathbf{w}\|_0 \leq K$$

- an investor wishes to hold only a small subset of available assets (e.g., due to transaction costs or simplicity).

Turnover Constraint: Controls changes in asset holdings.

$$\|\mathbf{w} - \mathbf{w}_0\|_1 \leq u$$

where $\mathbf{w}_0$ is the currently held portfolio.

- institutional investors aim to maintain stable holdings over time.

Market-Neutral Constraint: Ensures beta neutrality.

$$\boldsymbol{\beta}^T \mathbf{w} = 0$$

- $\boldsymbol{\beta}$ represents the vector of factor loadings (e.g., stock betas with respect to the market).

Mean Variance Portfolio

Mean Variance Portfolio Implementation

Integrated Estimation and Optimization

Maximum Sharpe Ratio Regression

Considering trading costs

Estimation

- Markowitz (1952) framework assumes full knowledge of return distributions.
- Real-world investors make portfolio decisions under uncertainty.
- $\mathcal{F}_{t-1}$ denotes the available historical data up to time $t-1$.
- $\mathbf{r}_t$ follows a multivariate stochastic process with:
  - **Conditional mean:**

$$\boldsymbol{\mu}_t \triangleq \mathrm{E}\left[\mathbf{r}_t \mid \mathcal{F}_{t-1}\right]$$

  - **Conditional covariance:**

$$\boldsymbol{\Sigma}_t \triangleq \mathrm{Cov}\left[\mathbf{r}_t \mid \mathcal{F}_{t-1}\right] = \mathrm{E}\left[\left(\mathbf{r}_t - \boldsymbol{\mu}_t\right)\left(\mathbf{r}_t - \boldsymbol{\mu}_t\right)^T \mid \mathcal{F}_{t-1}\right]$$

- These moments $(\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t)$ play a crucial role portfolio optimization.

Parameter Estimation

- Independent and Identically Distributed (i.i.d.) Model:

$$\mathbf{r}_t = \boldsymbol{\mu} + \epsilon_t$$

  - $\boldsymbol{\mu} \in \mathbb{R}^N$ is the mean return vector.
  - $\epsilon_t \in \mathbb{R}^N$ is an i.i.d. noise process with zero mean and constant covariance $\boldsymbol{\Sigma}$.
- **Estimation:** $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ need to be estimated from T observations.
- **Basic Sample Estimators:**
  - Sample mean: $\hat{\boldsymbol{\mu}} = \frac{1}{T}\sum_{t=1}^{T} \mathbf{r}_t$
  - Sample covariance matrix: $\hat{\boldsymbol{\Sigma}} = \frac{1}{T-1}\sum_{t=1}^{T}(\mathbf{r}_t - \hat{\boldsymbol{\mu}})(\mathbf{r}_t - \hat{\boldsymbol{\mu}})^T$

Plug-in Portfolios

- Investors often estimate return moments first, then optimize portfolio weights.
- The "plug-in" estimator replaces unknown parameters with estimated values.

$$\widehat{w} = \frac{1}{\gamma}\widehat{\Sigma}^{-1}\widehat{\mu}$$

Limitations:

- Consistency of estimates does not guarantee good finite-sample performance.
- Plug-in portfolios perform poorly when N is large relative to T.
  - Where else?
- High dimensional settings exacerbate estimation errors, leading to suboptimal decisions.

Economic Loss from Estimation Error

In Kan and Zhou [2007], the difference in investor utility between the true optimal portfolio w^* and the plug-in portfolio $\widehat{w}$ is treated as an economic loss function:

$$\mathrm{E}\left[L\left(w^*, \widehat{w}\right) \mid \mu, \Sigma\right] = U\left(w^*\right) - \mathrm{E}[U(\widehat{w}) \mid \mu, \Sigma]$$

- Statistical "risk function"
- Utility loss from using the plug-in portfolio can be quantified:

$$\mathrm{E}\left[L\left(w^*, \widehat{w}\right) \mid \mu, \Sigma\right] = a_1 \frac{\mathrm{SR}^2}{2\gamma} + a_2$$

- Loss increases with the number of assets N and Sharpe ratio.
- Investors experience larger utility costs in high-dimensional settings.

Challenges in Parameter Estimation

- Estimation Accuracy:
  - The parameter estimates $\hat{\boldsymbol{\mu}}$ and $\hat{\boldsymbol{\Sigma}}$ are only reliable when T is large.
  - For small T, the estimation error becomes significant and unacceptable.
  - The sample mean is particularly inefficient and produces very noisy estimates (Meucci, 2005).
- Limitations on Large T: In practice, the number of observations T is limited due to:
  - Historical data may not be sufficiently long.
  - Financial data properties change over time. vs. IID assumption
- Impact on Portfolio Design:
  - Traditional mean-variance portfolios suffer from estimation errors, leading to poor performance.
- **Practical Relevance:** Due to these limitations, Markowitz portfolio optimization and its extensions are rarely used by practitioners.

Mitigating Plug-in Portfolio Issues

- Alternative approaches reduce estimation uncertainty:
  - Shrinkage estimators (Bayes-Stein, Ledoit-Wolf shrinkage).
  - Black-Litterman estimators
  - Other Bayesian and high-dimensional covariance estimators
  - **Portfolio constraints to limit impact of estimation error.**

- Integrated estimation and utility optimization improves performance.

Drawbacks of Markowitz Portfolio

1. Variance as a Risk Measure:
   - it penalizes both unwanted high losses and desired low losses.
   - Solution: Use alternative risk measures, such as:
     - Value at Risk (VaR)
     - Conditional Value at Risk (CVaR)
2. Sensitivity to Parameter Estimation Errors:
   - The portfolio is highly sensitive to errors in the estimation of $\boldsymbol{\Sigma}$ and $\boldsymbol{\mu}$ (especially problematic)
     - Estimation + Optimization
   - Solution: Use robust optimization techniques and improved parameter estimation.
3. Lack of Risk Diversification:
   - Markowitz optimization considers only the total portfolio risk but ignores diversification.
   - This leads to excessive concentration of risk in a few assets.
   - Solution: more balanced risk allocation.

Mean Variance Portfolio

Mean Variance Portfolio Implementation

Integrated Estimation and Optimization

Maximum Sharpe Ratio Regression

Considering trading costs

Quadratic utility + i.i.d. Normality

- Under simple assumptions (i.i.d. normal returns, quadratic utility), the optimal portfolio should depend on sample statistics:
$$\widehat{w} = f(\widehat{\mu}, \widehat{\Sigma})$$
- Kan and Zhou [2007] show that certain linear modifications improve the plug-in estimator.
  - mixing with plug-in minimum variance portfolio
- good choices for $\widehat{w}$ "can potentially be a very complex nonlinear function of $\widehat{\mu}$ and $\widehat{\Sigma}$ and there can be infinitely many ways to construct it. However, it is not an easy matter to determine the optimal $f(\widehat{\mu}, \widehat{\Sigma})$."

Integrated Estimation and Optimization

- Unknown function f can be parameterized flexibly, e.g., using neural networks.
- The goal is to optimize expected out-of-sample utility while managing estimation uncertainty.

Machine Learning Workflow for Portfolio Choice:

Step 1: Model Training

- Choose a function class indexed by a tuning parameter (e.g., ridge penalty z).
- Estimate model parameters using the training sample.

Step 2: Validation and Regularization

- Select tuning parameters that optimize expected out-of-sample performance.
- Use cross-validation to approximate expected utility without closed-form solutions.

Generalizing Portfolio Choice with Machine Learning

- The portfolio rule can be written as:

$$\widehat{w} = f_\theta\left(X_T\right)$$

- X_T can include past returns, macroeconomic indicators, and other conditioning variables.
- the return distribution need not be normal nor i.i.d.
- Flexible machine learning methods allow customization for tail risk, multi-horizon objectives, and alternative return distributions.

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OLS Regression for Portfolio Weights

- The Markowitz plug-in portfolio solution:

$$w^{\mathrm{OLS}} \propto \widehat{\Sigma}^{-1}\widehat{\mu}$$

- Proportional to an OLS regression coefficient:

$$1 = w' R_t + u_t$$

where u_t are noise.

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  where u_t are noise.
- Maximum Sharpe Ratio Regression (MSRR) *seeks a combination of the risky excess returns R_t that behaves as closely as possible to a positive constant.*
- tantamount to finding the optimal combination of assets that maximizes the Sharpe ratio.
- Regression formulation is attractive for incorporating machine learning methods into parameterized portfolio problems.

Example: LASSO in MSRR

$$\widehat{\boldsymbol{w}^*} = \arg\min_{\boldsymbol{w}} ||\widehat{\boldsymbol{r}_c} - \boldsymbol{R}\boldsymbol{w}||_2^2 \quad \text{subject to } ||\boldsymbol{w}||_1 \leq \lambda,$$

Table 6

Summary of risk, Sharpe ratio and p-value of Sharpe ratio test of the portfolios under comparison on DJIA 30 index constituents and Fama-French three factors. The testing period is February 1967 – December 2016. The risk constraint $\sigma = 0.0366$ is the standard deviation of the excess returns on DJIA 30 index during February 1962 – January 1967.

DJIA 30 & FF3	σ=0.0366	$T = 60$	
Portfolio	Risk	Sharpe Ratio	p-value
Index	0.044	0.102	0.000
Equally weighted	0.043	0.064	0.000
Factor	0.040	0.425	0.000
KZ	0.124	0.535	0.000
MAXSER	0.044	**0.701**	–

Dynamic Portfolio Weight Parameterization

- Portfolio weights parameterization:

$$\underbrace{w_t}_{N\times 1} = \underbrace{S_t}_{N\times K}\underbrace{\beta}_{K\times 1}$$

  where S_t is a set of K signals.
- This introduces dynamic adjustments based on conditioning signals.
- Substituting dynamic weights into MSRR:

$$1 = \beta'(S'_{t-1}R_t) + u_t = \beta' F_t + u_t$$

- This representation enables factor-based portfolio construction.
- from the space of base returns to the space of factors

- w_t as a set of dynamic weights on the N base assets R_t.
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- PCR
- Parametric portfolio: "reduces considerably the room for model mispecification and estimation error."
  - dimension reduction
  - relative to estimating all assets' means and covariances
- The estimated portfolio coefficients are $\widehat{\beta} = \widehat{\mathrm{Cov}}(F)^{-1}\widehat{\mathrm{E}}(F)$.
- Simple OLS test statistics at our disposal for evaluating the contribution of each signal

Advantages of MSRR Framework

- Machine learning-friendly formulation enabling:
  - Large K regressions incorporating rich conditioning signals.
    - full zoo of factor signals and perhaps their interactions with macroeconomic state variables,
  - Regularization techniques (LASSO, ridge, ENET) for estimation stability.
  - Adaptations for constraints like leverage control.
    - L_1 penalty

Example: Regularization in MSRR

- Penalized MSRR with ridge regression:

$$\min_{\beta} \sum_{t} (1 - \beta' F_t)^2 + z\beta'\beta$$

- Solution:

$$\widehat{\beta} \propto (\widehat{\mathrm{Cov}}(F) + zI)^{-1} \widehat{\mathrm{E}}(F)$$

- Shrinks portfolio estimates for better out-of-sample performance.

Relaxing the Identical Coefficient Restriction

$$w_{i,t} = s'_{i,t}\beta_i$$

- Associated regression representation:

$$1 = \text{vec}(B)' \left\{ \text{vec}(S_{t-1}) \odot (R_t \otimes 1_K) \right\} + u_t$$

- Finds the tangency portfolio among NK "managed" strategies.
- Penalizing $\beta_i - \bar{\beta}$ allows some heterogeneity across assets but shrinks all assets' weight rules toward the average rule, in the spirit of empirical Bayes.
- For more general utility functions, estimation must step out of the OLS regression framework and requires numerical optimization.

High Complexity MSRR with Neural Networks

- MSRR can be extended using machine learning techniques such as neural networks.
- Nonlinear transformations improve flexibility in portfolio construction.
- Signals:

$$S_{i,t}(\theta) = q\left(X_{i,t};\theta\right)$$

- Portfolio weights:

$$w_{i,t} = \beta' S_{i,t}(\theta)$$

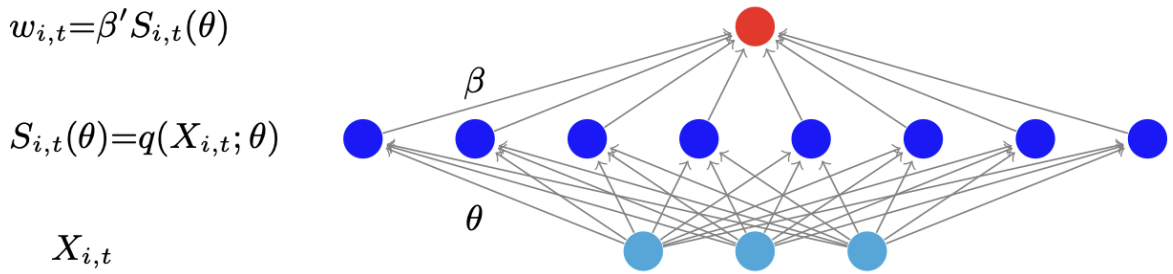

Figure 5.1: MSRR Neural Network

Note: Illustration of a simple neural architecture mapping conditioning variables $X_{i,t}$ into optimal portfolio weights $w_{i,t}$.

MSRR Neural Network Estimation

- The MSRR neural network formulation follows:

$$1 = \beta' S_{t-1}(\theta)' R_t + u_t$$

- Parameters (β, θ) are estimated via nonlinear least squares.
- This framework searches for optimal transformations of input signals to enhance Sharpe ratio strategies.

Expected Utility Framework

Simon et al. [2025] assume

- Investor maximizes expected utility of future portfolio returns:

$$\max_{\{w_{i,t}\}_{i=1}^{N_t}} E_t\left[u\left(r_{p,t+1}\right)\right] = E_t\left[u\left(\sum_{i=1}^{N_t} w_{i,t} r_{i,t+1}\right)\right].$$

- Portfolio weights depend on firm characteristics $x_{i,t}$ and estimated return distributions.
- Unlike standard models, this framework excludes a risk-free asset.
- Focus is placed on dynamic allocation among risky assets.
- Weights $w_{i,t}$ dynamically adjust based on market information at time t.

Estimation of Portfolio Weights

- Portfolio weights follow a parametric form:

$$w_{i,t} = f\left(x_{i,t}; \theta\right)$$

- The parameter vector θ is constant across assets and periods.
- θ is estimated by maximizing unconditional expected utility:

$$\max_{\theta} \frac{1}{T} \sum_{t=1}^{T} u\left(\sum_{i=1}^{N_t} f\left(x_{i,t}; \theta\right) r_{i,t+1}\right).$$

Empirical Evidence:

- Simon et al. [2025] implement a neural MSRR with a feed-forward architecture.
- Neural portfolio improves Sharpe ratio from 1.8 (linear model) to 2.5.

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Raymond Kan and Guofu Zhou. Optimal portfolio choice with parameter uncertainty. *Journal of Financial and Quantitative Analysis*, 42(3):621–656, 2007.

Frederik Simon, Sebastian Weibels, and Tom Zimmermann. Deep parametric portfolio policies. Technical report, CFR Working Paper, 2025.